



Permutations & Combinations

What Are Permutations & Combinations?

Permutations and combinations are ways to arrange or choose things from a group. They help us count possibilities!

Why learn this? To solve fun problems with counting! From picking team members to arranging books on a shelf.



Key Terms to Know



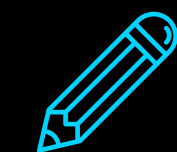
Permutation

Arrangement where order matters.
Example: ABC is different from CBA!



Combination

Selection where order doesn't matter.
Example: Choosing friends for a team.



Factorial (!)

Multiply numbers down to 1. Example: $3! = 3 \times 2 \times 1 = 6$



Understanding Permutations

A permutation is an arrangement where **ORDER MATTERS**. Imagine arranging 3 different colored beads on a necklace - Red, Blue, and Green can be arranged in many different ways!

Permutation Example:

3 Beads

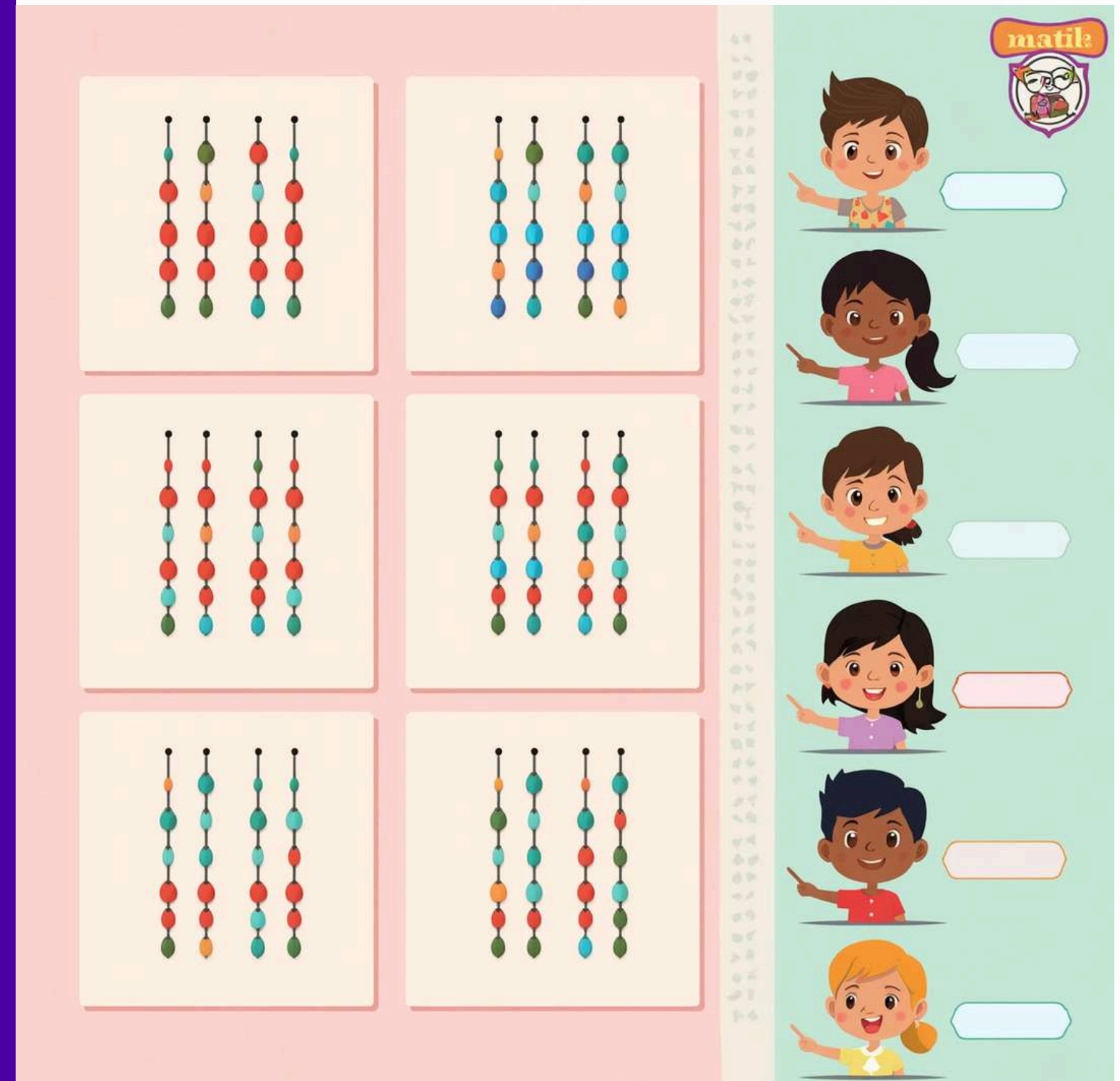
How many ways can we arrange Red, Blue, and Green beads?

Calculation: $3! = 3 \times 2 \times 1 = 6$ ways

All 6 arrangements:

RBG, RGB, BRG, BGR, GRB, GBR

Each order creates a unique pattern!





Understanding Combinations

In combinations, order does NOT matter! Example: Choosing 2 fruits from Mango, Banana, and Apple to make a salad. Picking Mango + Banana is the same as Banana + Mango!

Combination

Example: 3 Fruits

How many ways to choose 2 fruits from
Mango, Banana, Apple?

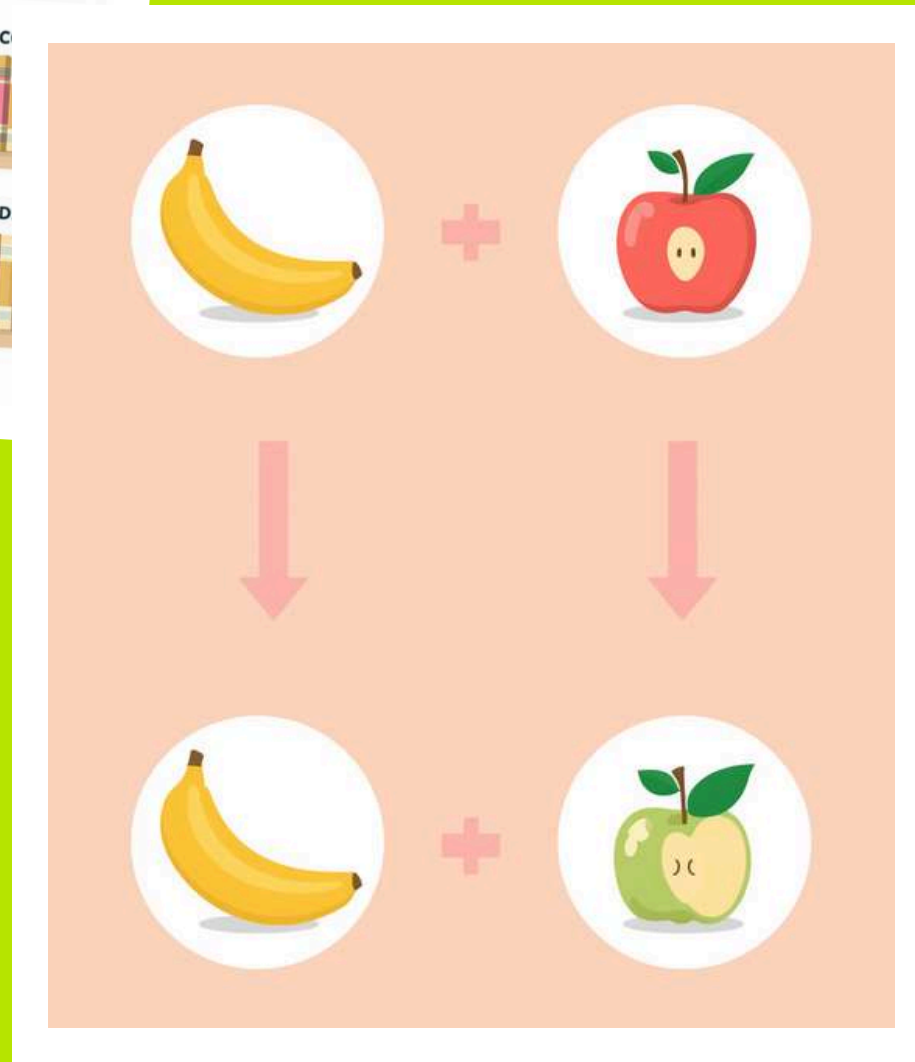
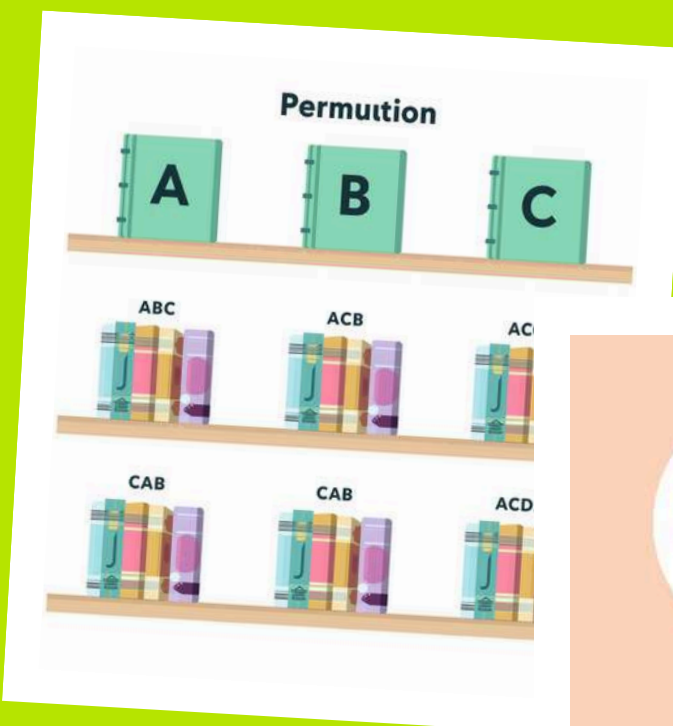
Calculation: $3 \text{ choose } 2 = 3 \text{ ways}$

The 3 pairs are:

- Mango + Banana
- Mango + Apple
- Banana + Apple



• Permutations vs Combinations



- Permutations: Order MATTERS**
- Arranging 3 books = 6 different ways
 - Each arrangement is unique
- Combinations: Order DOESN'T matter**
- Choosing 2 fruits from 3 = only 3 ways
 - Same items, fewer groups!

Real-Life Examples

in India



Arranging Diyas

How many ways can you arrange 4 colorful diyas in a row for Diwali decoration?



Cricket Team Selection

Choose 5 players from 8 friends to form your gully cricket team!



Festive Sweet Box

Pick 3 sweets from laddoo, barfi, gulab jamun, and jalebi for your mithai box!

Practice Problems



- 1.
- 2.
- 3.
- 4.

- 1.
- 2.
- 3.
- 4.

How many ways can you arrange 4 different colored flags in a row?
In how many ways can you pick 3 books from a shelf of 5 books?

How many ways can you arrange 3 diyas in a line for Diwali decoration?
You have 4 ladoos. In how many ways can you choose 2 to share with friends?



Solutions to Practice Problems

Problem 1: 4 flags = $4! = 24$ ways

Problem 2: Choose 3 from 5 books = 10 ways

Problem 3: Arrange 3 diyas = 6 ways

Quick Quiz

Test Your Understanding



Question 1

True or False: In permutations, the order of items matters.



Question 2

Choosing 2 toppings for your dosa from 4 options is a: A) Permutation B) Combination



Question 3

What is $4!$ (4 factorial)? A) 12 B) 24 C) 16

Summary of What We Learned



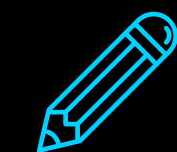
Permutations

Arrangements where ORDER matters.
Use factorial ($n!$) to calculate.



Combinations

Selections where order does NOT matter.
Fewer ways than permutations.



Real-Life Uses

Arranging diyas, picking cricket teams,
choosing sweets - math is everywhere!

Keep Exploring!

Look for permutations and combinations
in your daily life - from arranging books
on your shelf to choosing snacks!

Play math games and puzzles to practice.
Stay curious and never stop asking
questions - that's how mathematicians grow!





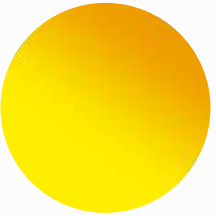
Thank You! for Learning

Thank you for learning Permutations & Combinations with us!
We hope you enjoyed exploring the fun world of counting and arranging.



Questions? Ask your teacher!
Keep practicing math every day
You are amazing mathematicians! 🌟





Fun Math

Fact! Bonus Slide



Did you know? The great Indian mathematician Bhaskara II discovered many counting principles over 800 years ago! His work on permutations helps us solve math puzzles even today. Math truly connects us across time!

